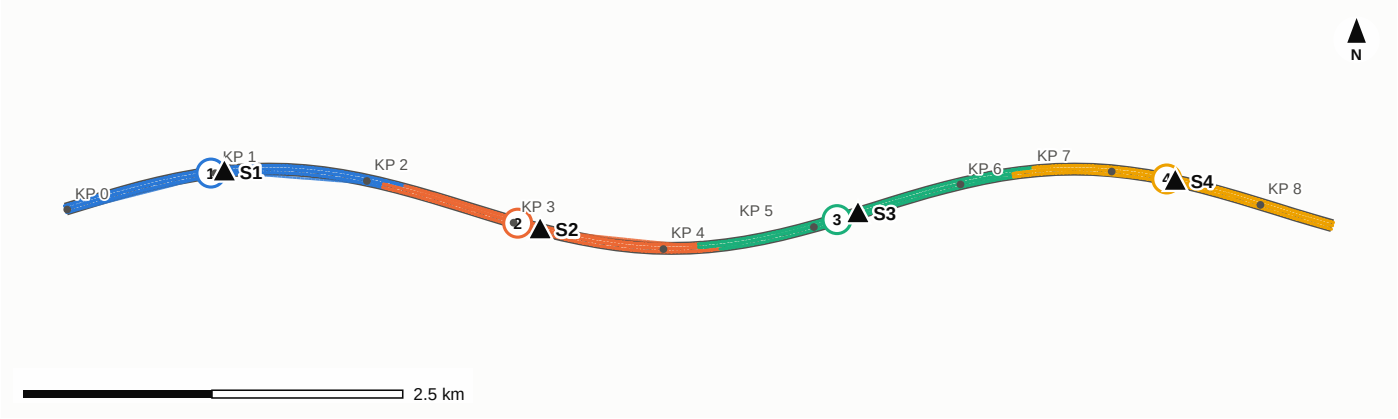
# Gas pipeline KP 0-8.5 corridor survey

Drone mission field brief · DJI Matrice 350 RTK + P1, 11 Oct 2026

## Summary

| | | | |
|---|---|---|---|
| Aircraft<br>**DJI Matrice 350 RTK + P1** | Mission<br>**route** | Area<br>**59.39 ha** | Route<br>**8.48 km, 4 sections** |
| Height<br>**88 m** | Return-to-home height<br>**88 m** | Battery-change height<br>**88 m** | GSD<br>**1.10 cm/px** |
| Speed<br>**5.5 m/s** | Photos<br>**1866** | Flights<br>**4** | Flight time<br>**1h 51m** |
| Usable / battery<br>**33.8 min** | Wind<br>**10 kn (5.0 m/s) from 250°** | Temp / payload<br>**24 °C / 0.00 kg** | Accessory power<br>**0 W** |
| On site<br>**07:00 – 10:00 (3.0 h)** | Battery sets<br>**6 (recommended 6)** | Charge waiting<br>**0 min** | |

## Mission map

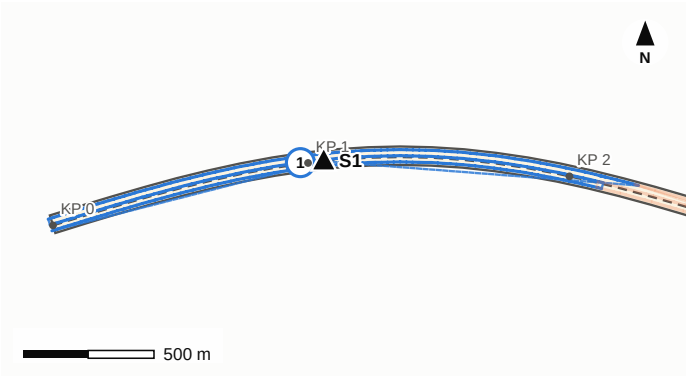



Flight 1 Flight 2 Flight 3 Flight 4 Transit to / from launch station Launch station

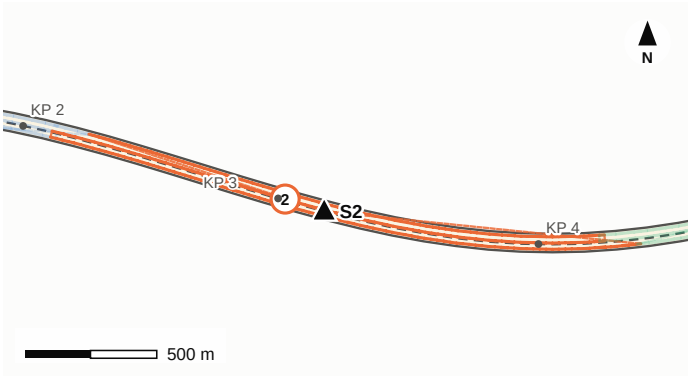
Grid north. Map and plan in the mission's planning coordinate system.

## Section maps

### S1 · flight 1 · KP 0.00–2.26

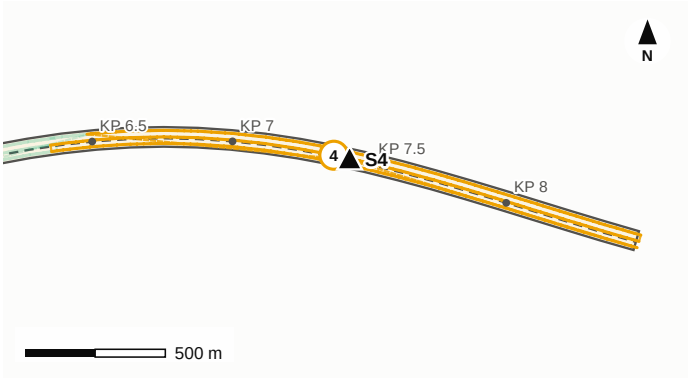



### S2 · flight 2 · KP 2.11–4.38



**S3 · flight 3 · KP 4.23–6.50**

**S4 · flight 4 · KP 6.35–8.48**




## Launch stations

| Station | Lat, Lon (WGS84) | Flights launched |
|---|---|---|
| **S1** | -32.276346, 115.747320 | 1 |
| **S2** | -32.279983, 115.769243 | 1 |
| **S3** | -32.279244, 115.791366 | 1 |
| **S4** | -32.277540, 115.813465 | 1 |

## Flights & battery plan

| # | Launch -> land | Set | Start | End | Min | Battery | Photos | KP (km) | Notes | Done |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | S1 -> S1 | 1 | 07:15 | 07:43 | 28.5 | 96% | 480 | 0.00–2.26 | battery change mid-line. | ☐ |
| 2 | S2 -> S2 | 2 | 08:00 | 08:27 | 27.8 | 91% | 466 | 2.11–4.38 | battery change mid-line. drive 12 min | ☐ |
| 3 | S3 -> S3 | 3 | 08:44 | 09:12 | 27.9 | 96% | 467 | 4.23–6.50 | battery change mid-line. drive 12 min | ☐ |
| 4 | S4 -> S4 | 4 | 09:28 | 09:55 | 27.0 | 88% | 453 | 6.35–8.48 | drive 12 min | ☐ |

## Terrain (DEM)

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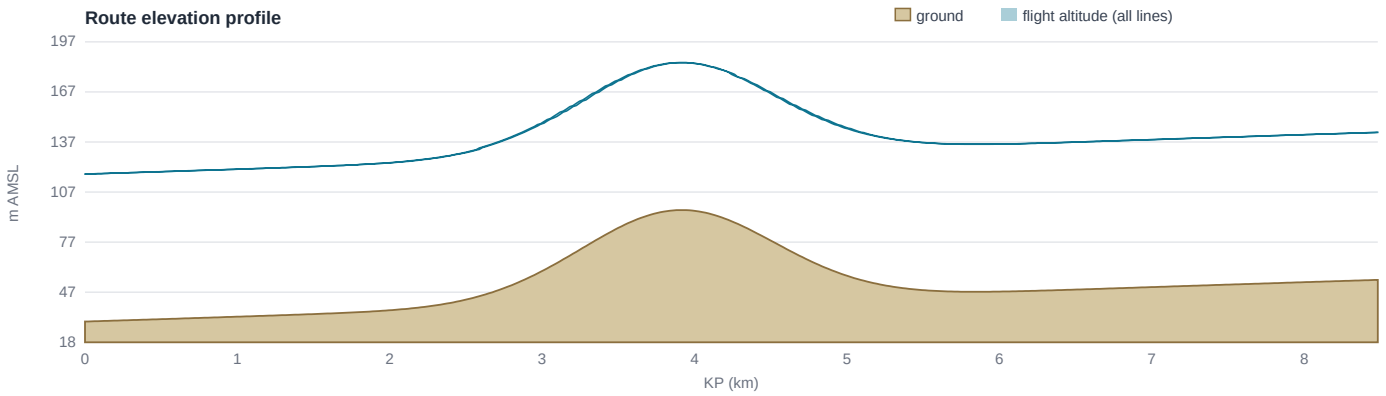
Terrain & obstacle check:
  Lowest clearance  : 88 m  (limit 30 m)
  Highest above gnd : 91 m  (limit 120 m)
  GSD over terrain  : 1.10 - 1.10 cm/px (mean 1.10)
  Height over survey: 88 m - 88 m above ground
  Worst overlap     : 75% front / 74% side (where ground is highest)
  OK - no clearance or height-limit problems found.
```




## Warnings & checks

- Check airspace, NOTAMs, weather and the mission preview in the flight app before each flight.